

Trigonometric Identities

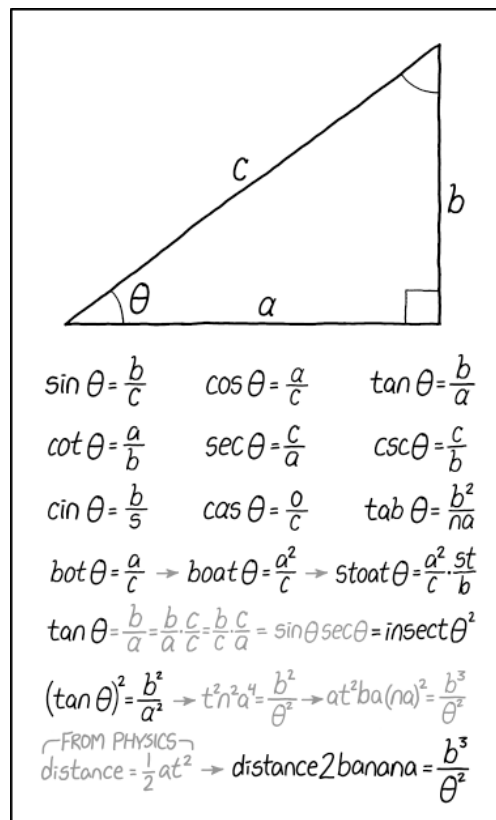
Expectations

Students will be able to:

- Prove trigonometric identities.
- State equivalent expressions.

Homework Log

Lesson	Date	Section	Assigned Work
1		5.5 Trig Identities Day 1	Text: pg. 310 #2 – 12
2		5.5 Trig Identities Day 2	LP: pg. 5
3		5.5 Trig Identities Day 3	LP: pg. 6 – 7



KEY TRIGONOMETRIC IDENTITIES

Figure 1 https://imgs.xkcd.com/comics/trig_identities.png

5.5 Trigonometric Identities

An algebraic statement such as $4x - 3x = x$ is true for all values of x and is called an **identity**. The following are examples of identities:

$$2 + 3 = 5$$

$$3(x - 7) = 3x - 21$$

$$\sin \theta = \frac{o}{H}$$

$$\cos \theta = \frac{A}{H}$$

$$\tan \theta = \frac{o}{A}$$

It is not always obvious that both sides of an equation are identical and so the fun begins with the proof.

We can use identities we already know to prove new identities.

Reciprocal Identities

$$1. \csc \theta = \frac{1}{\sin \theta}$$

$$2. \sec \theta = \frac{1}{\cos \theta}$$

$$3. \cot \theta = \frac{1}{\tan \theta}$$

Quotient Identities

$$4. \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$5. \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

6. $\sin^2 q + \cos^2 q = 1$

7. $1 + \tan^2 q = \sec^2 q$

8. $1 + \cot^2 q = \csc^2 q$

Note: These identities can be rearranged.

eg. Since $\tan x = \frac{\sin x}{\cos x}$, then $\cos x = \frac{\sin x}{\tan x}$ or, since $\sin^2 x + \cos^2 x = 1$, then $\sin^2 x = 1 - \cos^2 x$ and $\cos^2 x = 1 - \sin^2 x$.

Strategies that may help prove trigonometric identities:

1. Express functions in terms of $\sin x$ and $\cos x$.

Prove $\sin \theta \sec \theta \cot \theta = 1$

2. Look for squares and use the Pythagorean identities.

Prove that $\sin^6 \alpha = (1 - \cos^2 \alpha)^3$

3. Algebraic manipulations of distribution (expanding), factoring, and creating common denominators may allow cancellations in order to get to simpler expressions.

a) Prove that $(\sin \beta + \cos \beta)^2 + (\sin \beta - \cos \beta)^2 = 2$.

b) Prove that $\frac{\sin^2 x}{1 - \cos x} = 1 + \cos x$.

c) Prove that $\tan x + \frac{\cos x}{1 + \sin x} = \sec x$.

The most important strategy of all is patience, perseverance and ingenuity. If you get stuck, start again with a different outlook and plan. Good Luck!

Trigonometric Identities Practice

1. State an equivalent expression for each of the following.

- | | | | | |
|------------------------------------|-------------------------|--------------------------|--------------------------------|------------------------------------|
| a) $\sin^2 \theta$ | b) $\cos^2 \theta$ | c) $\tan^2 \theta$ | d) $\csc^2 \theta$ | e) $\sec^2 \theta$ |
| f) $\cot^2 \theta$ | g) $1 - \sin^2 \theta$ | h) $\sec^2 \theta - 1$ | i) $1 - \csc^2 \theta$ | j) $\sec^2 \theta - \tan^2 \theta$ |
| k) $\cot^2 \theta - \csc^2 \theta$ | l) $\frac{1}{\sec^2 q}$ | m) $\frac{-1}{\csc^2 q}$ | n) $\frac{\cos^2 q}{\sin^2 q}$ | |

Prove the following identities.

2. $\cos q' \tan q = \sin q$

3. $\frac{\cot q}{\tan q} = \frac{1 - \sin^2 q}{1 - \cos^2 q}$

4. $\cot^2 q = \frac{\cos^2 q}{1 - \cos^2 q}$

5. $\frac{\csc x}{\sec x} = \cot x$

6. $(\sin q + \cos q)^2 = 1 + 2 \sin q \cos q$

7. $2 \sin^2 q - 1 = \sin^2 q - \cos^2 q$

8. $\frac{1}{\sin^2 q} + \frac{1}{\cos^2 q} = \frac{1}{\sin^2 q \cos^2 q}$

9. $\cos^2 x = \sin^2 x + 2 \cos^2 x - 1$

10. $\tan q = \tan^2 q' \cot q$

11. $\sec^2 q + \csc^2 q = \sec^2 q' \csc^2 q$

12. $\frac{1}{1 + \sin q} + \frac{1}{1 - \sin q} = 2 \sec^2 q$

13. $\tan^2 x - \sin^2 x = \sin^2 x' \tan^2 x$

14. $\cot^2 q + \sec^2 q = \tan^2 q + \csc^2 q$

15. $\frac{1 + 2 \sin x \cos x}{\sin x + \cos x} = \sin x + \cos x$

16. $\frac{\sec q + 1}{\sec q - 1} + \frac{\cos q + 1}{\cos q - 1} = 0$

17. $\frac{\tan x \sin x}{\tan x + \sin x} = \frac{\tan x - \sin x}{\tan x \sin x}$

18. $\frac{\csc q + \cot q}{\csc q - \cot q} = \frac{1 + 2 \cos q + \cos^2 q}{\sin^2 q}$

19. $\sin q' \cos q' \tan q = 1 - \cos^2 q$

20. $\sin q + \tan q = \tan q(1 + \cos q)$

21. $\tan x + \cot x = \sec x \csc x$

22. $(1 - \cos^2 x)(1 + \cot^2 x) = 1$

23. $\frac{\csc q}{\csc q - 1} + \frac{\csc q}{\csc q + 1} = 2 \sec^2 q$

24. $\sec^4 q - \tan^4 q = 1 + 2 \tan^2 q$

25. $\sec^2 q - \sin^2 q = \cos^2 q + \tan^2 q$

26. $\sin^4 q - \cos^4 q = 1 - 2 \cos^2 q$

27. $\frac{\cos q}{\sec q} - \frac{\sin q}{\cot q} = \frac{\cos q \cot q - \tan q}{\csc q}$

Determine which of the following are identities.

28. $\cot q + \cos q = \tan q + \sin q$

29. $\cot q + \cos q = \frac{\cos q(1 + \sin q)}{\sin q}$

30. $\cot q + \cos q = \frac{2 \cos q}{\sin q}$

Fundamental Trigonometric Identities		
Reciprocal Identities	Quotient Identities	Pythagorean Identities
$\csc x = \frac{1}{\sin x}$	$\tan x = \frac{\sin x}{\cos x}$	$\sin^2 x + \cos^2 x = 1$
$\sec x = \frac{1}{\cos x}$	$\cot x = \frac{\cos x}{\sin x}$	$\sec^2 x = 1 + \tan^2 x$
$\cot x = \frac{1}{\tan x}$		$\csc^2 x = 1 + \cot^2 x$

Exercise / Homework:

1. Write in terms of $\sin x$ and $\cos x$, then simplify if possible.

a) $\csc x + \sec x$

b) $\tan x + \sec x$

c) $\cot x + \sin x$

d) $\tan x \cos x$

2. Factor each of the following.

a) $\sin^2 x + \sin^2 x \cos^2 x$

b) $1 - \cos^2 x$

c) $\tan^2 x - \cot^2 x$

d) $\sin^2 x - 2 \sin x + 1$

3. Simplify.

a) $\sin x \left(\frac{1}{\cos x} \right)$

b) $(\cos x)(\sec x)$

c) $1 - \cos^2 x$

d) $1 - \sin^2 x$

e) $\frac{\tan x}{\sin x}$

f) $(1 - \sin x)(1 + \sin x)$

g) $\left(\frac{1}{\tan x} \right) \sin x$

h) $\frac{1 + \tan^2 x}{\tan^2 x}$

i) $\frac{\sin x \cos x}{1 - \sin^2 x}$

j) $\frac{1 - \cos^2 x}{\sin x \cos x}$

4. Prove the following identities.

a) $\sin^2 x (1 + \cot^2 x) = 1$

b) $1 - \cos^2 x = \tan x \cos x \sin x$

c) $\cos x \tan^3 x = \sin x \tan^2 x$

d) $1 - 2 \cos^2 \theta = \sin^4 \theta - \cos^4 \theta$

e) $\cot \sigma + \frac{\sin \sigma}{1 + \cos \sigma} = \csc \sigma$

f) $\frac{\sec \varphi}{\sin \varphi} + \frac{\csc \varphi}{\cos \varphi} = \frac{2}{\sin \varphi \cos \varphi}$

g) $\sin^4 x + 2 \cos^2 x - \cos^4 x = 1$

h) $\sec x (1 - \cos x) = \sec x - 1$

i) $\frac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin x \cos x} = 1 - \tan x$

j) $\frac{1}{1 + \cos x} + \frac{1}{1 - \cos x} = 2 \csc^2 x$

*k) $1 - (\sin^6 x + \cos^6 x) = 3 \sin^2 x \cos^2 x$

l) $\sin^2 \alpha + 2 \cos^2 \alpha - 1 = \cos^2 \alpha$

(*most difficult)

Answers:

1. a) $\frac{1}{\sin x} + \frac{1}{\cos x}$ b) $\frac{\sin x + 1}{\cos x}$ c) $\frac{\cos x + \sin^2 x}{\sin x}$ d) $\sin x$ 2. a) $\sin^2 x (1 + \cos^2 x)$ b) $(1 - \cos x)(1 + \cos x)$ c) $(\tan x - \cot x)(\tan x + \cot x)$
d) $(\sin x - 1)^2$ 3. a) $\tan x$ b) 1 c) $\sin^2 x$ d) $\cos^2 x$ e) $\sec x$ f) $\cos^2 x$ g) $\cos x$ h) $\csc^2 x$ i) $\tan x$ j) $\tan x$

Trig Identities More Practice

Quotient Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{-\cos \theta}{\sin \theta}$$

Reciprocal Identities

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$\cot^2 \theta + 1 = \csc^2 \theta$$

1. Prove that each of the following equations is an identity by simplifying the left or right side of the equation.

a) $\cot x \sin x = \cos x$

b) $\sec \theta (1 - \cos \theta) = \sec \theta - 1$

c) $\sin^2 \beta (1 + \cot^2 \beta) = 1$

d) $\frac{1 + \tan \alpha}{1 + \cot \alpha} = \tan \alpha$

e) $\tan \theta + \cot \theta = \sec \theta \csc \theta$

f) $\sin^2 x \sec^2 x = \sec^2 x - 1$

g) $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1$

h) $\sin^2 x + \tan^2 x = \sec^2 x - \cos^2 x$

i) $(\cos x - \sin x)^2 = 1 - 2 \sin x \cos x$

j) $\frac{1}{\sec x + \tan x} = \sec x - \tan x$

k) $\frac{\tan^2 A}{1 + \tan^2 A} = \sin^2 A$

l) $\frac{\cos \theta}{1 - \sin \theta} - \sec \theta = \tan \theta$

m) $\frac{2 \sin^2 x - 1}{\sin x \cos x} = \tan x - \cot x$

n) $\sin^4 x - \cos^4 x = 1 - 2 \cos^2 x$

o) $\frac{\cot x}{\csc x - 1} + \frac{\cot x}{\csc x + 1} = 2 \sec x$

Challenge: Prove the following identities:

a) $\frac{1}{1 + \sin x} = \sec^2 x - \frac{\tan x}{\cos x}$

b) $\frac{\tan^3 x}{1 + \tan^2 x} + \frac{\cot^3 x}{1 + \cot^2 x} = \frac{1 - 2 \sin^2 x \cos^2 x}{\sin x \cos x}$